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Coordinating the Construction of Singapore MRT

by Govindan Sridharan, CCE

INTRODUCTION

The Singapore Mass Rapid Transit System comprises of 42 stations (15 underground and 27 elevated) distributed along a 67 km route. It is a "heavy" twin track railway system with an international standard gauge of 1,435 mm.

The geographical shape of the railway is that of an inverted "T" with routes linking the residential areas in the north central, west and east of the island to the business and commercial centre in south central part. The railway is served by one main depot in the north-south line and two sidings for overnight storage with minor repair facilities near the east and west end of the railway. Figure 1 shows the layout in detail.

IMPLEMENTATION

The final decision to build the railway system was made by the Government of Singapore in late 1982. The Mass Rapid Transit Corporation (MRTC) was formed with the specific task of building the system. The initial decision was to build the system in two phases over a period of nine years. The first contracts that went out for tender in 1984 received very enthusiastic and competitive response from the international contractors possibly due to the severe downturn in the construction industry during that period. The Government quickly reevaluated its earlier decision to build each phase separately. Even though the corporation would be required to increase an already massive recruitment program so that both the phases could be adequately supervised simultaneously, the advantages

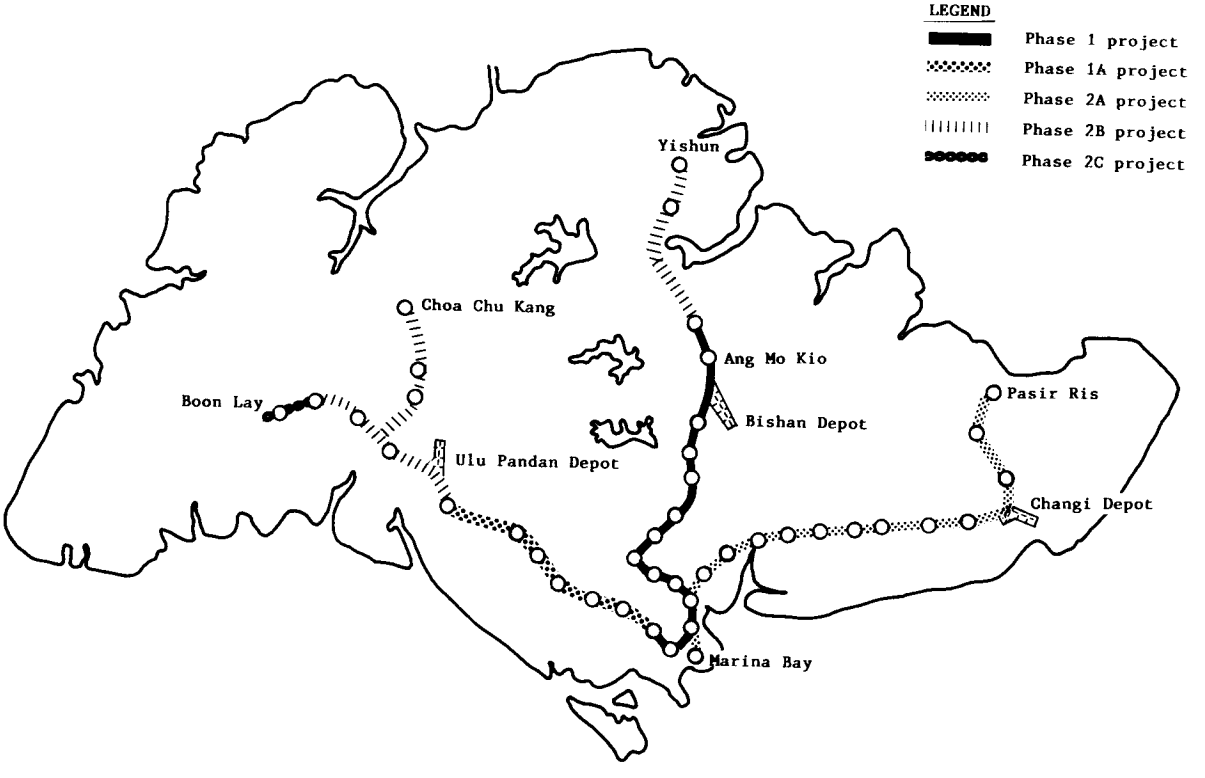


Figure 1 -- System Map

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of lower prices for the contracts and advancing the revenue service by two years were overwhelming.

With the Government's approval, MRTC launched its ambitious program in 1985 to complete the construction of the system by 1990. At the peak of its construction, 33 major civil contractors and 9 systemwide electrical and mechanical contractors were working over the entire system, being supervised by a project division staff of 1,500. The railway was divided into 9 sections for the purpose of opening the system to public. The passenger service in the first section was introduced on 7 Nov '89 followed by the official opening of the system comprising 20 stations along a 29 km of route by the prime minister Lee Kuan Yew in March 1988. By the end of 1989, 38 stations over a 59 km route were completed and opened to the public. The remaining four stations will be opened to passenger service by June 1990 right on the accelerated schedule and well within the S$5 billion budget.

The reasons for the resounding success of the implementation are many. However this paper will deal with one of the key elements essential for any project to succeed -- the coordination of interdisciplinary design and construction.

PROJECT COORDINATION MODEL

An effective and comprehensive communication system is a necessary ingredient for any coordination effort to succeed. The corporation had the envious task of building up its own management team while trying to get on with the design and construction of the system as early as possible. This necessitated awarding the first phase of the contracts on a design and construct basis. This method of contracting had the obvious advantage of utilising the civil contractor's project management team to take on the design and construction coordination with the interfacing systemwide E&M contractors.

The second phase of the project was built using the architect/engineer's design. However, the corporation continued with the practice of entrusting the coordination responsibility with the civil contractor as this method worked satisfactorily in the first phase. The coordination model adopted for the project is illustrated in Figure 2.

DESIGN COORDINATION

The general specification of the contracts provided clear guidelines regarding design coordination. Specific dates by which design inputs and outputs from the interfacing contractors were required were specified. The contractors were required to establish a coordination team and the project coordinator was identified as a key staff member. The coordination team must also have an effective representation on site during construction.

Despite the precautions taken, design coordination proved to be a difficult task for the following reasons:

- the international contractors had little or no experience in the local environment.
- the simultaneous development of E&M equipment design with the civil design resulted in changes to combined services drawing even beyond the design freeze dates. 10.
- the overlapping of construction and design resulted in continuous pressure to complete the design with incomplete information.

The corporation's own coordination team had no choice but to involve itself in the coordination process in order to minimise abortive coordination. This was done through structured regular coordination meetings between affected parties to resolve conflicts and allow coherent development of the integrated design.

The coordination model did not envisage such a direct involvement and the coordinating contractor could construe this as an extra to his contract. However, all the parties involved recognised the

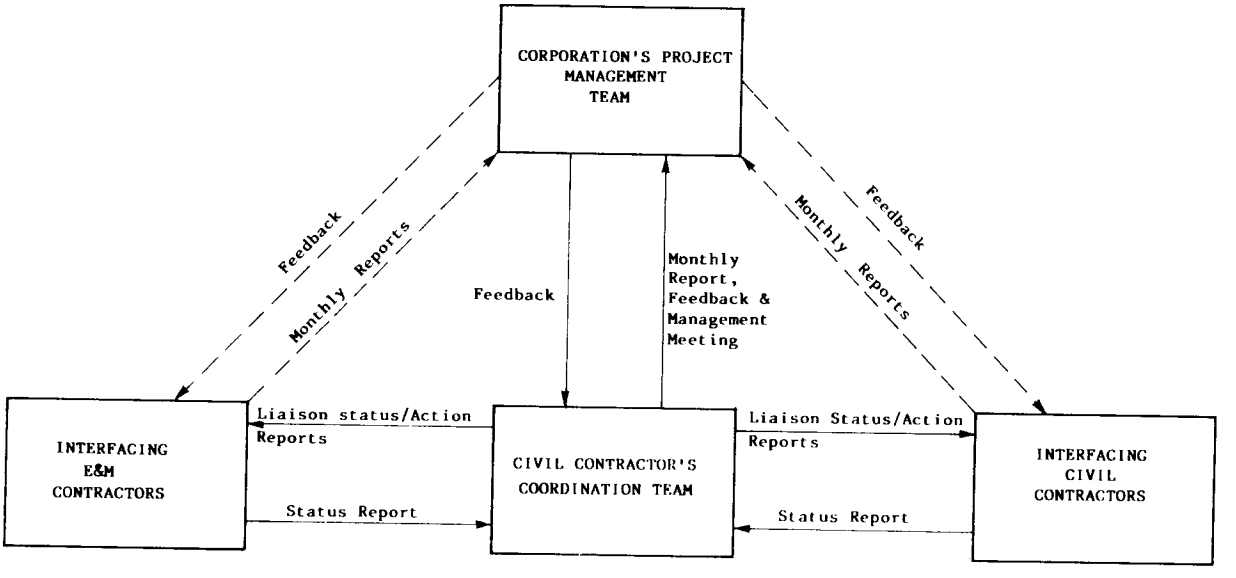


Figure 2 -- Project Coordination Model

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positive contribution made by the corporation's project management team in cutting down response time and very few claims resulted.

CONSTRUCTION COORDINATION

The civil contractor had dual construction coordination responsibilities. First, he had to produce combined services drawings (CSD's) for the station, incorporating the design information from systemwide contractors into his architectural and structural layout. These drawings show the principal service routes of all systems. In parallel with the CSD's, sets of drawings were produced depicting the structural interfaces with the E&M systems. These drawings, known as structural E&M drawings (SEM's), show all significant penetrations, plinths, curbs etc which affect structural design. Both the drawings were produced from the preliminary CSD's and SEM's prepared by the architect/engineer.

The second coordination responsibility was the preparation of a coordinated installation program. An orderly establishment of priorities of access within agreed time periods to each room in the station is the primary aim of this program. The corporation envisaged that unless specific guidelines were provided in the contract, the civil contractor would find it extremely difficult to produce a meaningful program. It was also foreseen that the systemwide E&M contractors would find it frustrating to coordinate with 33 civil contractors if no common definitions regarding the level of completion of architectural finishes before accessing the specific rooms were not available.

The contract specified the degrees of architectural finishes for each major area in the station and also provided a preliminary coordinated installation program for each area (See Table 1 and Figure 3). The civil contractor used this program and prepared the sequence of installation works in specific areas through to commissioning in the form of a time based bar chart and circulated the program to the respective System-wide contractors for comments. The iterative process of refining the program continued through formal meetings and a final program signed off by all contractors working in the station was submitted to the corporation for approval. The corporation specified in the contract, the key date by which such iterative process should be completed. This was usually three months before the actual planned access date by the first systemwide contractor to the station.

The formal coordinated installation program accepted by all the contractors and the corporation was used to monitor the progress on site. The detailed day to day construction coordination was carried through 4-week rolling programs prepared in simple bar chart forms. This required a lot of effort from all concerned parties, but the resulting program was extremely useful in resolving conflicts. The site meetings were conducted by the civil contractor and attended by the E&M contractors with the corporation's planning staff attending only as observers. The Corporation intervened to resolve conflicts only on an exception basis.

CONCLUSION

The overall coordination process in this project proved to be an exhausting affair to all parties involved. The cooperation extended by the contractors and the corporation's timely intervention to sort out any differences among interfacing contractors were mainly responsible for the successful completion of the project. The same spirit of cooperation cannot be taken for granted in every project. The following recommendations are made to the future builders of mass rapid railway systems in the area of coordination.

Table 1 --Project Coordination Model

LOCATION - Escalator zone

DEGREE 1	DEGREE 2	DEGREE 3
1. Complete basic structure.	1. Install false ceiling (if any) during escalator stop-work period.	1. Install escalator cladding.
2. Escalator walls, M/C and Halon rooms complete, clean, dry and waterproof.	2. Install bulkhead during escalator stop-work period.	2. Install advertising panels.
3. Access for truss available.	3. Temporary lockable doors.	3. Permanent doors/locks.
4. Wall finishes complete.	4. Steel ladder (if any).	4. Completion of: a. Light fittings/terminations. b. Small power fittings and terminations. c. Fire protection/detection system.
5. Floor screed adjacent to top/bottom landing complete.	5. Completion of all first/second fix* of electrical/mechanical/fire protection items.	

Note: First fix items include, but are not limited to following: Conduits, cable trays, trunking, junction boxes, socket outlets, pipework and ducting.

Second fix items include, but are not limited to following: Wiring, cabling, sockets, insulation, cladding and painting.

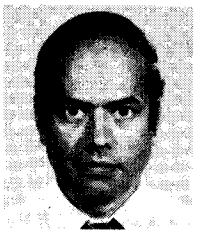
1. Particular attention should be paid at the beginning of the project to the packaging of the scope of work. It may be preferable to hire a specialist designer to design the station as a single system including E&M works such as air conditioning, lighting and elevators. This will avoid time-consuming coordination effort during construction as this package can be let as a single construction package. However, specialised works such as trackwork, signalling and communication may have to be designed and constructed on a systemwide basis.

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2) It may be preferable that the coordination effort between systemwide contractors and station builders are carried out by the owner through a dedicated coordination team.

3) E&M equipment design and layout should be completed before embarking on finalising the architectural layout of the stations.

4) The key operational and maintenance staff should be recruited at the beginning of the project. Their input at the system planning stage will result in a rational design and practical routing of services.



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FIG. 4 PRELIMINARY COORDINATED INSTALLATION PROGRAMME FOR PASIR RIS STATION

REF NO	SPECIFIED PART OF WORK (YR. / NTH.)	1989 JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	1990 JAN	FEB	MAR
	MILESTONES			↓ BASIC STRUCTURE				↓ WORKS TRAIN			↓ POWER ON				↓ TEST RUNNING	↓ TRIAL RUNNING
1	PLATFORM PUBLIC AREA										▽ DEG 2				▽ DEG 3	
2	ALL PUBLIC AREAS OTHER THAN PLATFORM AND ENTRANCE										▽ DEG 2				▽ DEG 3	
3	ESCALATOR ZONE, ESCALATOR MACHINE RM	▽ DEG 1									▽ DEG 2				▽ DEG 3	
4	LIFT SHAFT LIFT MACHINE ROOM			▽ DEG 2											▽ DEG 3	
5	COMMUNICATION EQUIPMENT RM, RELAY RM							▽ DEG 2							▽ DEG 3	
6	POWER PLANT RMS							▽ DEG 2			▽ DEG 3					
7	A/C CONTROL RM										▽ DEG 2				▽ DEG 3	
8	A/C PLANT RM, PUMP RM, EJECTOR RM										▽ DEG 2				▽ DEG 3	
9	STATION CONTROL RM			▽ ACCESS FOR SCR CONTRACTOR												
10	AFC RM AND TICKET RM, TICKET AND CASH RM										▽ DEG 2				▽ DEG 3	

Figure 1 -- Preliminary Coordinated Installation Programme for Pasir Ris Station

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